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Inventor(s)	Wilcox; Robert et al.

Three dimensional LED device and method of manufacture

Abstract

Solid-state lighting devices, and more particularly to a three-dimensional (3D) light-emitting diode (LED) device and a method of manufacture are disclosed. A submount of the LED can have several submount portions that are angled with respect to each other, and LED chips can be mounted on each submount portion, such that the LED chips of the device are angled at different angles with respect to each other. In an embodiment, the LED device can include a heatsink that is in contact with each of the submount portions to channel heat away from the LED chips. The LED chips can be mounted on the submount portions when all submount portions are laying flat, and then the submount portions can be pushed or punched into their respective angles.

Inventors: Wilcox; Robert (Rolesville, NC), Check; Michael (Holly Springs, NC), Blakely; Colin (Raleigh, NC), Suich; David (Durham, NC), Sokol; Joseph G. (Durham, NC), Pertuit; Andre (Raleigh, NC)

Applicant: CreeLED, Inc. (Durham, NC)

Family ID: 92545631

Assignee: CreeLED, Inc. (Durham, NC)

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Primary Examiner: Dzierzynski; Evan P

Attorney, Agent or Firm: Withrow & Terranova, P.L.L.C.

Background/Summary

RELATED APPLICATIONS (1) This application is a division of U.S. patent application Ser. No. 18/198,339, filed May 17, 2023, now U.S. Pat. No. 12,078,300, the disclosure of which is hereby incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates to a light-emitting diode (LED) device and more particularly to a three-dimensional LED device and method for manufacturing the same.

BACKGROUND

(2) Solid-state lighting devices such as light-emitting diodes (LEDs) are increasingly used in both consumer and commercial applications. Advancements in LED technology have resulted in highly efficient and mechanically robust light sources with a long service life. Accordingly, modern LEDs have enabled a variety of new display applications and are being increasingly utilized for general illumination applications, often replacing incandescent and fluorescent light sources.

(3) LEDs are solid-state devices that convert electrical energy to light and generally include one or more active layers of semiconductor material (or an active region) arranged between oppositely doped n-type and p-type layers. When a bias is applied across the doped layers, holes and electrons are injected into the one or more active layers where they recombine to generate emissions such as visible light or ultraviolet emissions. An LED chip typically includes an active region that may be fabricated, for example, from gallium nitride, gallium phosphide, aluminum nitride, gallium arsenide-based materials, and/or from organic semiconductor materials. Photons generated by the active region are initiated in all directions.

(4) LED packages have been developed that can provide mechanical support, electrical connections, and encapsulation for LED emitters. Lumiphoric materials, such as phosphors, may also be arranged in close proximity to LED emitters to convert portions of light emissions to different wavelengths. As LED technology continues to be developed for ever-evolving modern applications, challenges exist in keeping up with operating demands for LED packages and related elements of LED packages.

(5) LED packages that are designed to provide wide area illumination generally include a number of LED devices mounted in different orientations such that the LED chips in the LED devices emit light in different directions. The LED chips are conventionally oriented in an LED device in a single orientation, thus requiring multiple LED devices to achieve wide area light distribution of greater than 180 degrees.

(6) The art continues to seek improved LEDs and solid-state lighting devices having desirable illumination characteristics capable of overcoming challenges associated with conventional lighting devices.

SUMMARY

(7) The present disclosure relates to solid-state lighting devices, and more particularly to a three-dimensional (3D) light-emitting diode (LED) device and a method of manufacture. A submount of the LED can have several submount portions that are angled with respect to each other, and LED chips can be mounted on each submount portion, such that the LED chips of the device are angled at different angles with respect to each other. In an embodiment, the LED device can include a heatsink that is in contact with each of the submount portions to channel heat away from the LED chips. The LED chips can be mounted on the submount portions when all submount portions are laying flat, and then the submount portions can be pushed or punched into their respective angles.

(8) In an embodiment, an LED device can include a submount comprising a plurality of submount portions that have respective surfaces that are angled with respect to each other and with respect to a surface of a remainder of the submount and an LED chip that is mounted on each of the respective surfaces of the plurality of submount portions.

(9) In an embodiment, the plurality of submount portions comprise electrical traces upon which the plurality LED chips are mounted.

- (10) In an embodiment the plurality of submount portions are not in direct contact with each other.
- (11) In an embodiment the LED device can include a heat sink in contact with each of the plurality of submount portions.
- (12) In an embodiment the LED device can include an LED chip mounted on a top surface of the heat sink with a normal that is parallel to the surface of the remainder of the submount.
- (13) In an embodiment the LED device can include a cover structure covering the plurality of submount portions and LED chips.
- (14) In an embodiment each submount portion of the plurality of submount portions comprise a plurality of LED chips mounted thereon.
- (15) In an embodiment each LED chip of the plurality of LED chips is configured to emit a different color of light.
- (16) In an embodiment the plurality of submount portions are at least one of a triangular or quadrilateral shape.
- (17) In an embodiment a method of fabricating an LED device includes mounting a plurality of LED chips on a top surface of a plurality of submount portions of a submount, wherein the plurality of submount portions are coplanar with respect to a remainder of the submount and inserting a punch through the submount between the plurality of submount portions, wherein in response to the inserting the punch, the plurality of submount portions are angled with respect to each other and to the remainder of the submount.
- (18) In an embodiment, the method includes scoring one or more edges of the plurality of submount portions prior to inserting the punch.
- (19) In an embodiment, the plurality of LED chips are mounted on electrical traces that flex in response to inserting the punch.
- (20) In an embodiment, the method includes controlling a distance through which the punch is inserted through the submount based on a predetermined angle of the plurality of submount portions with respect to the remainder of the submount.
- (21) In an embodiment, the method includes placing a heat sink between the plurality of submount portions, such that the heatsink is in contact with a surface of each submount portion of the plurality of submount portions.
- (22) In an embodiment, the punch comprises a heatsink at a tip of the punch, and the method further includes detaching the heatsink from the punch, wherein the heatsink is in contact with, and connected to surfaces of the plurality of submount portions and retracting the punch.
- (23) In an embodiment, the method includes inserting the punch through a bottom surface of the plurality of submount portions, causing a top surface of plurality of submount portions to be angled away from the punch.
- (24) In an embodiment the method includes inserting the punch through a top surface of each of the plurality of submount portions, one at a time, causing a top surface of plurality of submount portions to be angled towards the punch.
- (25) In an embodiment, each submount portion of the plurality of submount portions comprises a plurality of LED chips mounted thereon.
- (26) In an embodiment, each LED chip of the plurality LED chips is configured to emit a different color light.
- (27) In an embodiment, the plurality of submount portions are at least one of a triangular or quadrilateral shape.
- (28) In another aspect, any of the foregoing aspects individually or together, and/or various separate aspects and features as described herein, may be combined for additional advantage. Any of the various features and elements as disclosed herein may be combined with one or more other disclosed features and elements unless indicated to the contrary herein.
- (29) Those skilled in the art will appreciate the scope of the present disclosure and realize

additional aspects thereof after reading the following detailed description of the preferred embodiments in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

- (1) The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure, and together with the description serve to explain the principles of the disclosure.
- (2) FIG. 1A is a cross-sectional view of a light-emitting diode (LED) device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.
- (3) FIG. 1B is a top down view of LED device of FIG. 1A according to aspects disclosed herein.
- (4) FIG. 2 is a cross-sectional view of another LED device with a plurality of LED chips that are angled with respect to each other at a different angle than shown in FIG. 1A according to aspects disclosed herein.
- (5) FIG. 3 is a cross-sectional view of an LED device similar to FIG. 1A with a heatsink according to aspects disclosed herein.
- (6) FIG. 4 is a cross-sectional view of an LED device similar to FIG. 3 with an additional LED chip mounted on the heatsink according to aspects disclosed herein.
- (7) FIG. 5A is a cross-sectional view of another LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.
- (8) FIG. 5B is a top down view of the LED device of FIG. 5A according to aspects disclosed herein.
- (9) FIG. 6 is a cross-sectional view of an LED device similar to FIG. 5A with a heatsink according to aspects disclosed herein.
- (10) FIG. 7 is a cross-sectional view of another LED device with a plurality of LED chips on each submount portion that are angled with respect to LED chips on other submount portions according to aspects disclosed herein.
- (11) FIG. 8 is a cross-sectional view of another LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.
- (12) FIGS. 9A-9E depict a series of illustrations showing a method of forming an LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.
- (13) FIGS. 10A-10C depict a series of illustrations showing another method of forming an LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.
- (14) FIG. 11 is a flowchart of a method for forming an LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.

DETAILED DESCRIPTION

- (15) The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.
- (16) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from

the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(17) It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

(18) Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

(19) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(20) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(21) Embodiments are described herein with reference to schematic illustrations of embodiments of the disclosure. As such, the actual dimensions of the layers and elements can be different, and variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are expected. For example, a region illustrated or described as square or rectangular can have rounded or curved features, and regions shown as straight lines may have some irregularity. Thus, the regions illustrated in the figures are schematic and their shapes are not intended to illustrate the precise shape of a region of a device and are not intended to limit the scope of the disclosure. Additionally, sizes of structures or regions may be exaggerated relative to other structures or regions for illustrative purposes and, thus, are provided to illustrate the general structures of the present subject matter and may or may not be drawn to scale. Common elements between figures may be shown herein with common element numbers and may not be subsequently re-described.

(22) The present disclosure relates to solid-state lighting devices, and more particularly to a three-dimensional (3D) light-emitting diode (LED) device and a method of manufacture. A submount of the LED can have several submount portions that are angled with respect to each other, and LED chips can be mounted on each submount portion, such that the LED chips of the device are angled at different angles with respect to each other. In an embodiment, the LED device can include a

heatsink that is in contact with each of the submount portions to channel heat away from the LED chips. The LED chips can be mounted on the submount portions when all submount portions are laying flat, and then the submount portions can be pushed or punched into their respective angles. (23) Before delving into specific details of various aspects of the present disclosure, an overview of various elements that may be included in exemplary LED packages is provided for context. An LED chip typically comprises an active LED structure or region that can have many different semiconductor layers arranged in different ways. The fabrication and operation of LEDs and their active structures are generally known in the art and are only briefly discussed herein. The layers of the active LED structure can be fabricated using known processes with a suitable process being fabrication using metal organic chemical vapor deposition. The layers of the active LED structure can comprise many different layers and generally comprise an active layer sandwiched between n-type and p-type oppositely doped epitaxial layers, all of which are formed successively on a growth substrate. It is understood that additional layers and elements can also be included in the active LED structure, including, but not limited to, buffer layers, nucleation layers, super lattice structures, un-doped layers, cladding layers, contact layers, and current-spreading layers and light extraction layers and elements. The active layer can comprise a single quantum well, a multiple quantum well, a double heterostructure, or super lattice structures.

(24) The active LED structure can be fabricated from different material systems, with some material systems being Group III nitride-based material systems. Group III nitrides refer to those semiconductor compounds formed between nitrogen (N) and the elements in Group III of the periodic table, usually aluminum (Al), gallium (Ga), and indium (In). Gallium nitride (GaN) is a common binary compound. Group III nitrides also refer to ternary and quaternary compounds such as aluminum gallium nitride (AlGa_N), indium gallium nitride (InGa_N), and aluminum indium gallium nitride (AlInGa_N). For Group III nitrides, silicon (Si) is a common n-type dopant and magnesium (Mg) is a common p-type dopant. Accordingly, the active layer, n-type layer, and p-type layer may include one or more layers of GaN, AlGa_N, InGa_N, and AlInGa_N that are either undoped or doped with Si or Mg for a material system based on Group III nitrides. Other material systems include silicon carbide (SiC), organic semiconductor materials, and other Group III-V systems such as gallium phosphide (GaP), gallium arsenide (GaAs), and related compounds. The active LED structure may be grown on a growth substrate that can include many materials, such as sapphire, SiC, aluminum nitride (AlN), and GaN, with a suitable substrate being a 4H polytype of SiC, although other SiC polytypes can also be used including 3C, 6H, and 15R polytypes.

(25) Different embodiments of the active LED structure can emit different wavelengths of light depending on the composition of the active layer and n-type and p-type layers. In some embodiments, the active LED structure emits blue light with a peak wavelength range of approximately 430 nanometers (nm) to 480 nm. In other embodiments, the active LED structure emits green light with a peak wavelength range of 500 nm to 570 nm. In other embodiments, the active LED structure emits red light with a peak wavelength range of 600 nm to 650 nm. In certain embodiments, the active LED structure may be configured to emit light that is outside the visible spectrum, including one or more portions of the ultraviolet (UV) spectrum. The UV spectrum is typically divided into three wavelength range categories denoted with letters A, B, and C. In this manner, UV-A light is typically defined as a peak wavelength range from 315 nm to 400 nm, UV-B is typically defined as a peak wavelength range from 280 nm to 315 nm, and UV-C is typically defined as a peak wavelength range from 100 nm to 280 nm.

(26) An LED chip can also be covered with one or more lumiphoric materials (also referred to herein as lumiphors), such as phosphors, such that at least some of the light from the LED chip is absorbed by the one or more lumiphors and is converted to one or more different wavelength spectra according to the characteristic emission from the one or more lumiphors. In this regard, at least one lumiphor receiving at least a portion of the light generated by the LED source may re-emit light having different peak wavelength than the LED source. An LED source and one or more

lumiphoric materials may be selected such that their combined output results in light with one or more desired characteristics such as color, color point, intensity, etc. In certain embodiments, aggregate emissions of LED chips, optionally in combination with one or more lumiphoric materials, may be arranged to provide cool white, neutral white, or warm white light, such as within a color temperature range of from 1800K to 10,000K. In certain embodiments, lumiphoric materials having cyan, green, amber, yellow, orange, and/or red peak wavelengths may be used. In some embodiments, the combination of the LED chip and the one or more lumiphors (e.g., phosphors) emits a generally white combination of light. The one or more phosphors may include yellow (e.g., YAG:Ce), green (e.g., LuAg:Ce), and red (e.g., Ca.sub.i-x-ySr.sub.xEu.sub.yAlSiN.sub.3) emitting phosphors, and combinations thereof.

(27) Lumiphoric materials as described herein may be or include one or more of a phosphor, a scintillator, a lumiphoric ink, a quantum dot material, a day glow tape, and the like. Lumiphoric materials may be provided by any suitable means, for example, direct coating on one or more surfaces of an LED, dispersal in an encapsulant material configured to cover one or more LEDs, and/or coating on one or more optical or support elements (e.g., by powder coating, inkjet printing, or the like). In certain embodiments, lumiphoric materials may be downconverting or upconverting, and combinations of both downconverting and upconverting materials may be provided. In certain embodiments, multiple different (e.g., compositionally different) lumiphoric materials arranged to produce different peak wavelengths may be arranged to receive emissions from one or more LED chips. One or more lumiphoric materials may be provided on one or more portions of an LED chip in various configurations. In certain embodiments, one or more surfaces of LED chips may be conformally coated with one or more lumiphoric materials, while other surfaces of such LED chips may be devoid of lumiphoric material.

(28) Light emitted by the active layer or region of an LED chip is initiated in all directions. For directional applications, internal mirrors or external reflective surfaces may be employed to redirect as much light as possible toward a desired emission direction. Internal mirrors may include single or multiple layers. Some multi-layer mirrors include a metal reflector layer and a dielectric reflector layer, wherein the dielectric reflector layer is arranged between the metal reflector layer and a plurality of semiconductor layers.

(29) As used herein, a layer or region of a light-emitting device may be considered to be “transparent” when at least 80% of emitted radiation that impinges on the layer or region emerges through the layer or region. Moreover, as used herein, a layer or region of an LED is considered to be “reflective” or embody a “mirror” or a “reflector” when at least 80% of the emitted radiation that impinges on the layer or region is reflected. In some embodiments, the emitted radiation comprises visible light such as blue and/or green LEDs with or without lumiphoric materials. In other embodiments, the emitted radiation may comprise nonvisible light. For example, in the context of GaN-based blue and/or green LEDs, silver (Ag) may be considered a reflective material (e.g., at least 80% reflective). In the case of ultraviolet (UV) LEDs, appropriate materials may be selected to provide a desired, and in some embodiments high, reflectivity and/or a desired, and in some embodiments low, absorption. In certain embodiments, a “light-transmissive” material may be configured to transmit at least 50% of emitted radiation of a desired wavelength.

(30) The present disclosure can be useful for LED chips having a variety of geometries, such as vertical geometry or lateral geometry. A vertical geometry LED chip typically includes anode and cathode connections on opposing sides or faces of the LED chip. A lateral geometry LED chip typically includes both anode and cathode connections on the same side of the LED chip that is opposite a substrate, such as a growth substrate. In certain embodiments, a lateral geometry LED chip may be arranged for flip-chip mounting on another surface.

(31) According to aspects of the present disclosure, LED packages may include one or more elements, such as lumiphoric materials, encapsulants, light-altering materials, lenses, and electrical contacts, among others, that are provided with one or more LED chips. In certain aspects, an LED

package may include a support member, such as a submount or a lead frame. Suitable materials for a submount include, but are not limited to, ceramic materials such as aluminum oxide or alumina, AlN, or organic insulators like polyimide (PI) and polyphthalamide (PPA). In other embodiments a submount may comprise a printed circuit board (PCB), sapphire, Si, or any other suitable material. For PCB embodiments, different PCB types can be used such as standard FR-4 PCB, metal core PCB, or any other type of PCB. In still further embodiments, the support structure may embody a lead frame structure.

(32) As used herein, light-altering materials may include many different materials including light-reflective materials that reflect or redirect light, light-absorbing materials that absorb light, and materials that act as a thixotropic agent. As used herein, the term “light-reflective” refers to materials or particles that reflect, refract, or otherwise redirect light. For light-reflective materials, the light-altering material may include at least one of fused silica, fumed silica, titanium dioxide (TiO₂), or metal particles suspended in a binder, such as silicone or epoxy. For light-absorbing materials, the light-altering material may include at least one of carbon, silicon, or metal particles suspended in a binder, such as silicone or epoxy. The light-reflective materials and the light-absorbing materials may comprise nanoparticles. In certain embodiments, the light-altering material may comprise a generally white color to reflect and redirect light. In other embodiments, the light-altering material may comprise a generally opaque or black color for absorbing light and increasing contrast.

(33) In certain applications, LED devices as disclosed herein may be well suited in closely-spaced array applications such automotive lighting, general lighting, and lighting displays. For exterior automotive lighting, multiple LED devices may be arranged under a common lens or optic to provide a single overall emission or emissions that are capable of changing between different emission characteristics. Changing emission characteristics may include toggling between high beam and low beam emissions, adaptively changing emissions, and adjusting correlated color temperatures (CCTs) that correspond with daytime and nighttime running conditions. In general lighting applications, LED devices as disclosed herein may be configured to provide modules, systems, and fixtures that are capable of providing one or more different emission colors or CCT values, such as one or more of warm white (e.g., 1800 Kelvin (K)-3000 K), neutral white (e.g., 3500 K-4500 K), and cool white (5000 K-6500 K). LED devices may also include monochromatic LEDs without phosphors that emit colors such as violet, blue, cyan, green, amber, red, and photored. For horticulture lighting applications, LED devices as disclosed herein may be arranged to provide modules, systems, and fixtures that are capable of changing between different emission characteristics that target various growth conditions of different crops.

(34) FIG. 1A is a cross-sectional view of a light-emitting diode (LED) device **100** with a plurality of LED chips **106-1**, **106-2**, and **106-3** (referred to as **106** when referring to them collectively) that are angled with respect to each other according to aspects disclosed herein. The LED chips **106** can be mounted on respective submount portions **102-1**, **102-2**, and **102-3** (referred to as **102** when referring to them collectively) that are also angled with respect to each other. In an embodiment, the submount portions **102** can be at least partially connected to a remainder of the submount **104**. In an embodiment, the LED chips **106** can be mounted on metallic electrical traces **108** that connect the submount portions **102** to the remainder of the submount **104**. The LED chips **106** can be mounted on the submount portions **102** when the submount portions **102** are flat, or parallel with respect to the remainder of the submount, and after the LED chips **106** are mounted, the submount portions **102** can be manipulated such that they are at an angle with respect to each other and to the submount **104**. Further details of how the submount portions **102** are manipulated, and how the LED device **100** is formed are discussed with respect to FIGS. 9A-9E.

(35) In an embodiment, the material of the submount **104** and submount portions **102** can flex when the submount portions **102** are manipulated. In other embodiments, the material of the submount **104** and submount portions **102** may tear, either partially, or completely. In an

embodiment, the metal of the electrical traces **108** can flex and maintain a connection between the electrical traces on the submount **104** and the submount portions **102**. In an embodiment, the submount portions **102** may be in contact with each other before manipulation into the 3D form, but after the submount portions **102** are raised relative to the remainder of the submount **104**, the submount portions **102** are not in contact with each other.

(36) In the embodiment depicted in FIG. **1A**, and FIG. **1B**, which is a top-down view of the embodiment in FIG. **1A**, there can be 4 submount portions **102-1**, **102-2**, **102-3**, and **102-4**, each with an LED chip mounted thereon via the metallic electrical traces **108**. It is to be appreciated that submount portion **102-4** and LED chip **106-4** are not visible in FIG. **1A** as they are behind the submount portion **102-2**. In the embodiment shown in FIGS. **1A** and **1B**, there can be 4 submount portions, each arranged symmetrically around a central location, but in other embodiments, there can be any number of submount portions, arranged in any manner, depending on the desired light emission pattern of the LED device **100**. For example, the four submount portions **102** instead of being arranged uniformly 360 degrees around a central point, in other embodiments, two or more submount portions can be arranged around 180 degrees in order to provide more targeted illumination.

(37) The normal **114-1**, **114-2**, and **114-3** of the submount portions **102-1**, **102-2**, and **102-3** can be adjusted based on the desired emission pattern. For example, the normals **114** can be adjusted so that the angle between the normal **114** and the normal of the submount **104** is small to provide just slightly broader illumination than if the LEDs **106** were flat mounted on the submount **104**. In other embodiments, the angles of the normals **114** with respect to the normal of the submount **104** can be up to 90 degrees, or even higher.

(38) In some embodiments, there can be a recess **103** between the submount portions **102**. The recess **103** can be formed by a portion of the submount **104** being removed prior to the submount portions **102** being manipulated to different angles. In other embodiments, depending on the shape of the submount portions **102**, the recess **103** can be formed when the submount portions **102** are lifted with respect to the remainder submount **104**.

(39) In certain embodiments, the LED device **100** may comprise a cover structure **110** that may in some embodiments comprise a conversion layer that comprises lumiphoric material, or the cover structure **110** may comprise lumiphoric material throughout the cover structure **110**. The cover structure **110** may be attached to the LED chip **106** or the submount **104** on which the LED chips **106** are attached via the submount portions **102** using, for example, a layer of transparent adhesive such as silicone. It is to be appreciated that in FIG. **1A**, the scale of the cover structure **110**, and submount portions **102-1**, **102-2**, and **102-3** and LED chips **106-1**, **106-2**, and **106-3**, are not necessarily drawn proportionally, and that in other embodiments, the cover structure **110** may be much larger with respect to the submount portions and LED chips than is shown in FIG. **1A**. It is also to be appreciated that although the cover structure **110** is only shown with regard to the embodiment in FIG. **1A**, the cover structure **110** can also be included in any of the embodiments in the other figures, but has not been drawn for the sake of simplicity.

(40) FIG. **2** is a cross-sectional view of another LED device **100** with a plurality of LED chips **106** that are angled with respect to each other at a different angle than shown in FIG. **1A** according to aspects disclosed herein.

(41) In the embodiment shown in FIG. **2**, the submount portions **102** are oriented at 90 degrees with respect to the remainder of the submount portion **104**. In this embodiment, the normal **114** of the submount portions **102** are in a plane that is parallel to the submount portion **104**. In other embodiments, other angles are possible.

(42) FIG. **3** is a cross-sectional view of an LED device similar to FIG. **1A** with a heatsink according to aspects disclosed herein. The embodiment in FIG. **3** is similar to that depicted in FIG. **1A** except for the addition of the heatsink **112** in the place of the recess **103**. The heatsink **112** can be pyramidal in shape, with sides that are in contact with a back side surface of the submount

portions **102** in order to conduct thermal energy away from the submount portions **102** in order to extend the operating life of the LED chips **106**. The heatsink can be metal, e.g., copper or aluminum, or some other material with appropriate thermal properties. Although not shown, the heatsink **112** may have radiative fins or other elements to disperse the thermal energy. The heatsink **112** may also be connected to other components external to the LED device **100** that can transport thermal energy away from the LED device **100**.

(43) FIG. **4** is a cross-sectional view of an LED device similar to FIG. **3** with an additional LED chip mounted on the heatsink according to aspects disclosed herein. In order to improve light emission directly opposite the LED device **100**, the heatsink can have mounted thereon another LED chip **106-5**. In an embodiment, the LED chip **106-5** can be powered by metal traces on the heatsink that are connected to metal traces on one of the submount portions, for example, submount portion **102-2** in the embodiment shown in FIG. **4**. In other embodiments, the metal traces **108** on the heatsink may be coupled to metal traces on the remainder of the submount **104**.

(44) FIG. **5A** is a cross-sectional view of another LED device with a plurality of LED chips **106** that are angled with respect to each other according to aspects disclosed herein. FIG. **5B** is a top-down view of the LED device **100** of FIG. **5A** according to aspects disclosed herein. The embodiment in FIGS. **5A** and **5B** is similar to that shown in FIGS. **1A** and **1B**, except that instead of being square or rectangular, the submount portions **102** in FIGS. **5A** and **5B** are triangular. The LED chips in FIGS. **5A** and **5B** are shown as being triangular too, except that in other embodiments, the LED chips **106** can be square or rectangular or any other shape, while still being mounted on triangular submount portions **102**.

(45) In the embodiment shown in FIGS. **5A** and **5B**, there are six submount portions but only three of which (**102-1**, **102-2**, and **102-3**) have LED chips **106** mounted thereon. The other submount portions (e.g., **105**) can also be raised relative to the remainder of the submount **104**, but not include LED chips **106**. Alternatively, in other embodiments, the submount portions **105** can include LED chips **106** mounted thereon.

(46) In an embodiment, due to the triangular shape of the submount portions **102**, there is no recess **103** when the submount portions are flat, before being raised, but a recess **103** can be formed when the submount portions **102** are raised.

(47) FIG. **6** is a cross-sectional view of an LED device similar to FIG. **5A** with a heatsink **112** according to aspects disclosed herein. The heatsink **112** in FIG. **6** can be similar to the heatsink in FIGS. **3** and **4**, except with a different pyramidal shape so that surfaces of the heatsink **112** can be in contact with a back surface of the submount portions **102**. Optionally, the heatsink **112** in FIG. **6** can also have a LED chip **106** placed on top of the surface, like the embodiment depicted in FIG. **4**.

(48) FIG. **7** is a cross-sectional view of another LED device with a plurality of LED chips **106** on each submount portion **102** that are angled with respect to LED chips **106** on other submount portions **102** according to aspects disclosed herein. In the embodiment shown in FIG. **7**, each submount portion **102** can include a plurality of LED chips **106** mounted thereon. In the embodiment shown in FIG. **7** there are four LED chips **106** mounted on each submount portion **102**, but in other embodiments, there can be two or more LED chips mounted on each submount portion. There can be different numbers of LED chips mounted on each submount portion in various embodiments. Also, each submount portion **102** can have LED chips of a single color, or a plurality of colors. For example, submount portion **102-2**, can include LED chips **106-1**, **106-2**, **106-3**, and **106-4**. In an example, each of the LED chips **106-1**, **106-2**, **106-3**, and **106-4** can be the same color, while LED chips on other submount portions are other colors. Alternatively, LED chip **106-1** could be white, LED chip **106-2** could be red, LED chip **106-3** could be blue, and LED chip **106-4** could be green.

(49) FIG. **8** is a cross-sectional view of another LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein. In the embodiment shown in FIG. **8**, the submount portions are angled so that the normals **114** of the submount portions

intersect at some location away from the LED device **100**. The submount portions can be angled so that the focal point is at some predetermined location depending on the desired emission pattern.

(50) FIGS. **9A-9E** depict a series of illustrations showing a method of forming an LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.

(51) In FIG. **9A**, the submount **104** can lay flat with no submount portions **102** being raised. The electrical traces **108** can then be placed on the surface of the submount **104**. In an embodiment, the submount portions **102** can be scored, or completely cut while still in the flattened position.

(52) As shown in FIG. **9B**, the LED chips (e.g., **106-1** and **106-2**) can be placed on parts of the submount **104** that will become the submount portions **102** once raised. In an embodiment, the scoring, which is optional, may occur after the LED chips **106** are mounted.

(53) FIG. **9C** depicts a punch **116** that can be pushed through the submount **104** to push the submount portions **102** into their raised positions. In an embodiment, the punch **116** can optionally include the heatsink **112** that can be left behind when the punch retracts.

(54) FIG. **9D** depicts the punch **116** when the punch is just beginning to punch through the submount **104** and the submount portions **102** are being raised into position. The end position of the punch **116**, which controls the angle of the submount portions, can be controlled based on a predetermined angle of the plurality of submount portions **102** with respect to the remainder of the submount **104**.

(55) In FIG. **9E**, the punch **116** can be retracted. In the embodiment where there is a heatsink on the tip of the punch, the heatsink **112** can be left behind, forming the embodiment similar in FIG. **3**. In other embodiments, where there is no heatsink on the tip of the punch **116**, the embodiment formed can be similar to the embodiment in FIG. **1A**. Alternatively, the heatsink **112** can be placed in the recess **103** after the punch **116** is retracted.

(56) The submount portions **102** can be scored, either completely or partially, in order to facilitate a clean separation between the submount portions **102** and the remainder of the submount **104**. While the punch **116** is pushing the submount portions **102** into position, the submount material can either flex or partially tear. The electrical traces **108** can also flex or bend while being manipulated.

(57) In other embodiments, the electrical traces **108** on the submount portion **102** may not touch the electrical traces **108** on the remainder of the submount **104** when the submount portions **102** are flat, but after being punched, the electrical traces **108** can be moved closer to each other such that an electrical connection is formed.

(58) FIGS. **10A-10C** depict a series of illustrations showing another method of forming an LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.

(59) The embodiments shown in FIGS. **10A-10C** are another way of achieving a 3D LED component where the LED chips are mounted in different orientations. In the embodiment depicted, a rod **118** can have a plurality of sides (four in this embodiment, each of which has LED chips (e.g., **106-1** and **106-2**) mounted on the electrical traces **108** on the surface of the rod **118**. Different portions of the rod **118** can be cut or separated along plane **117** to form 3D LED components as shown in FIG. **10B** that have the LED chips **106** mounted thereon, and as shown in FIG. **10C**, the 3D LED components can then be mounted on a submount **104** that already has electrical traces placed thereon. The electrical trace of the 3D LED component (e.g., **108-1**) can then form an electrical contact with the electrical trace **108-2** of the submount **104**.

(60) FIG. **11** is a flowchart of a method for forming an LED device with a plurality of LED chips that are angled with respect to each other according to aspects disclosed herein.

(61) The method can begin at step **1102** where the method includes mounting a plurality of LED chips on a top surface of a plurality of submount portions of a submount, wherein the plurality of submount portions are coplanar with respect to a remainder of the submount. In an embodiment, the plurality of LED chips are mounted on electrical traces that flex in response to inserting the

punch.

(62) At step **1104**, the method includes inserting a punch through the submount between the plurality of submount portions, wherein in response to the inserting the punch, the plurality of submount portions are angled with respect to each other and to the remainder of the submount. Before the punch is inserted, the edges of the plurality of submount portions can be scored.

(63) The end position of the punch **116**, which controls the angle of the submount portions, can be controlled based on a predetermined angle of the plurality of submount portions **102** with respect to the remainder of the submount **104**.

(64) A heat sink can be placed between the plurality of submount portions, such that the heatsink is in contact with a surface of each submount portion of the plurality of submount portions. In an embodiment, the method can further include detaching the heatsink from the punch, wherein the heatsink is in contact with, and connected to surfaces of the plurality of submount portions before retracting the punch.

(65) According to an embodiment, to form the embodiment shown in FIG. **1A** where the submount portions **102** are angled facing away from each other, the punch can be inserted through a bottom surface of the plurality of submount portions, causing a top surface of plurality of submount portions to be angled away from the punch.

(66) Alternatively, to form the embodiment shown in FIG. **8** where the submount portions **102** are angled facing each other, the punch can be inserted through a top surface of the plurality of submount portions, causing a top surface of plurality of submount portions to be angled towards the punch. In other embodiments, one or more punches can individually punch each submount portion of the plurality of submount portions, one at a time, so as to avoid damaging the submount or metallic electrical traces **108**. In other embodiments, each submount portion **102** of the plurality of submount portions **102** can be pulled from the top side, by a claw, in order to place the submount portion **102** in position as shown in FIG. **8**.

(67) It is contemplated that any of the foregoing aspects, and/or various separate aspects and features as described herein, may be combined for additional advantage. Any of the various embodiments as disclosed herein may be combined with one or more other disclosed embodiments unless indicated to the contrary herein.

(68) Those skilled in the art will recognize improvements and modifications to the preferred embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A method of fabricating a light-emitting diode (LED) device, comprising: mounting a plurality of LED chips on a top surface of a plurality of submount portions of a submount, wherein the plurality of submount portions are coplanar with respect to a remainder of the submount; and inserting a punch through the submount between the plurality of submount portions, wherein in response to the inserting the punch, the plurality of submount portions are angled with respect to each other and to the remainder of the submount.
2. The method of claim 1, further comprising scoring one or more edges of the plurality of submount portions prior to inserting the punch.
3. The method of claim 1, wherein the plurality of LED chips are mounted on electrical traces that flex in response to the inserting the punch.
4. The method of claim 1, further comprising: controlling a distance through which the punch is inserted through the submount based on a predetermined angle of the plurality of submount portions with respect to the remainder of the submount.
5. The method of claim 1, further comprising: placing a heat sink between the plurality of submount portions, such that the heatsink is in contact with a surface of each submount portion of

the plurality of submount portions.

6. The method of claim 1, wherein the punch comprises a heatsink at a tip of the punch, and the method further comprises: detaching the heatsink from the punch, wherein the heatsink is in contact with, and connected to surfaces of the plurality of submount portions; and retracting the punch.

7. The method of claim 1, wherein the inserting the punch comprises: inserting the punch through a bottom surface of the plurality of submount portions, causing a top surface of plurality of submount portions to be angled away from the punch.

8. The method of claim 1, wherein the inserting the punch comprises: inserting the punch through a top surface of each of the plurality of submount portions, one at a time, causing a top surface of plurality of submount portions to be angled towards the punch.

9. The method of claim 1, wherein each submount portion of the plurality of submount portions comprises a plurality of LED chips mounted thereon.

10. The method of claim 9, wherein each LED chip of the plurality LED chips is configured to emit a different color light.

11. The method of claim 1, wherein the plurality of submount portions are at least one of a triangular or quadrilateral shape.
